

. Net Core

C#

- 1) .NET !
- 2) .NET Framework !
- 3) Basics — DataTypes
- 4) OOP → Design Patterns + SOLID Principle
★ DRY Principle
- 5) Events and Delegates
- 6) Collections and Generic
- 7) File IO / Serialization
- 8) Reflection, Custom Attributes
- 9) ADO.NET → DB [MS SQL Server]

10) C# Features :-

⑩ → LINQ →

11) Threading

12) Exception Handling.

AspNet MVC Core Web Appl'n.

↳ MVC lifecycle.

higher languages

C++ code



gcc Compiler



.exe

[binary code]

Developer m/c

editor:

VS code

+
header files

packages

OS windows

[binary code]

app.exe ← d.c

+ packages

Client m/c

OS: windows

{
 Linux
 +
 loader
 +
 Assembler
}

OS

Windows
Runtime



m/c code

↳ CPU

Developer m/c

Java code ^{higher language}



javac compiler



.exe

[Byte code]

editor:

Eclipse, STS, IntelliJ
, vs code etc., Notepad++

+ JDK ✓

Client m/c

app.exe [Byte code] ^{↓ d.c}



Java Virtual Machine [JVM]



JIT [Just-In-Time Interpreter]

JDK



m/c code → CPU

Developer mlc

python code



python Compiler



.exe

[Byte code] → ATS

editor
VS code,
Jupyter

PyCharm
python
module

Client mlc

.exe [Byte code]

Python [Python Virtual Machine]

↓ → Interpreter

↓ mlc code → CPU

python module

Developer's m/c

.NET code



Compiler



.exe / .dll

[Dynamic link library]

editor :

Visual Studio 2022/2026
VS Code

Software Development Kit

[SDK] → .Net FW

[MSIL : Microsoft Intermediate language]

Client m/c

.exe ← [MSIL]



CLR (Common Language Runtime) ← SDK

+

JIT (Just-in-Time Compiler)



machine code → CPU

.NET Framework ? [SDK]

→ pre-compiled set of library files
(.dll files)

↳ pre-compiled MSIL

↓

Pre-JIT compiler

System.dll

System.Data.dll

System.Collections.dll

SDK editor Visual Studio

C#, Vb.Net, F#, J#,

Developer m/c
32 + lang. + 32 + compiler
ms2L

C# code



CSC



ms2L

.exe | .dll

Vb Net



VBC



ms2L

.exe | .dll

F#



FSC



ms2L

.exe | .dll

Client m/c

ms2L (.exe)

+ SDK

ms2L (.exe)

+ SDK

ms2L (.exe)

+ SDK

Common language Runtime (CLR)

JIT Compiler

m/c code → CPU

C++ lang

C#

int x = 10;

↓

CSC

↓

ms2L

Visual Basic

Vb.Net

Dim x as Integer

↓

VIBL

↓

ms2L

Rule :- Common Type System (CTS)

int	→	System.Int32
string	→	System.String
double	→	System.Double
short	→	System.Int16
bool	→	System.Boolean
char	→	System.Char

• .Net :- platform/P/W to develop
desktop, web appⁿ, Console
Command-line-Interface
(CLI), Web API's,
mobile, IoT appⁿ
etc etc

Standard .NET

Old .NET

OS: Windows

CLR:-

mscorlib.dll

JIT:-

- Pre-JIT compiler

- Normal / Standard JIT compiler

- Eco no -JIT ~~X~~

.NET Core

Open-source .NET

OS: Windows, Linux, MacOS

CLR:-

System.Private.CoreCLR.dll

JIT:-

Ryu-JIT compiler

. Net Core

C# / VB.NET



Roslyn Compiler



MSIL



↳ MSIL

↳ Ryu-JIT +

F#



FSC



MSIL



Core CLR + Ryu
↳ MIC → CPU

System.Private.
CoreCLR.dll

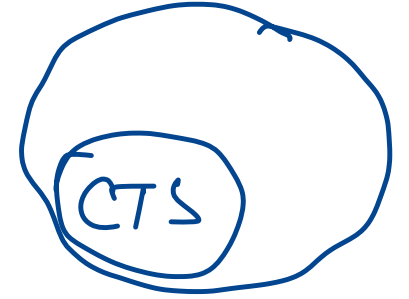
↳ MIC → CPU

Features of LLR :-

- 1) memory Allocation
- 2) memory De-allocation : Garbage collector [GC]
- 3) Security.
- 4) Exception handling.
- 5) JIT - compiler
- 6) m/c code $\rightarrow$ Threads $\rightarrow$ CPU
- 7) Loading dependencies.

Common Language Specifications [CLS]

Rules



- * OOP language
- * Every statement should end with ;
- * use 'new' keyword to create new objects of a class.
- * target CLR for → memory allocation
↳ GC for → memory de-allocation
- * Exception handling, security checks.
- * Should follow CTS Rules.

FCL
↓

Base Class Library Files
[BCL]

System.dll

System.Data.dll

System.Data.Xml.dll

System.Collections.dll

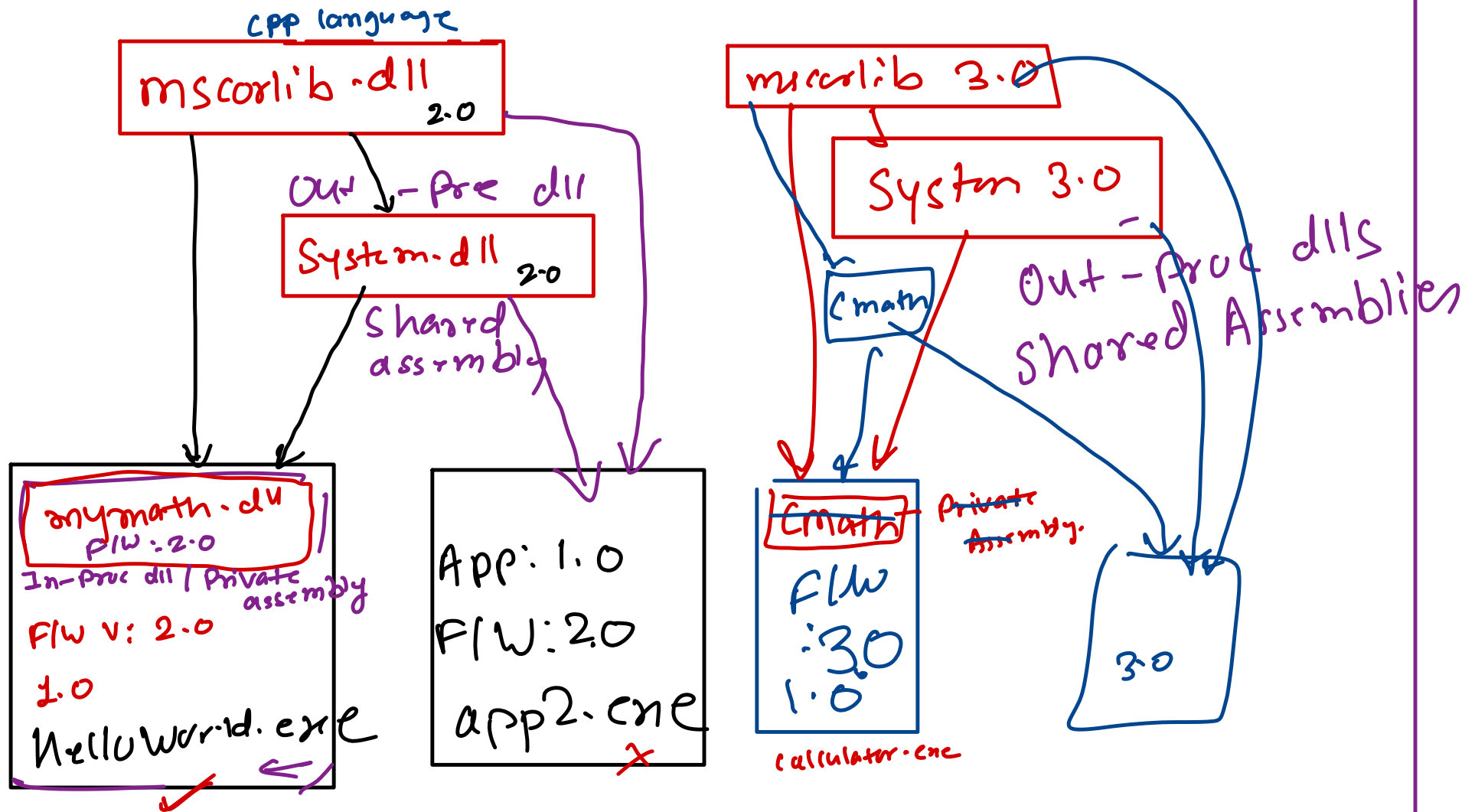
FW version : 1.0 | 2.0 | 3.0 | 3.5 | 4.0 | 4.7.

CLR version : 1.0 | 2.0 | ~~3.0~~ | 3.0 | 4.0 | 4.0

mscorlib.dll

C# version : 1.0 | 2.0 | 3.0 - - - C# 13

OS



cmath.dll
FW 3.0

myMath.dll → 2.0
→ 1.0

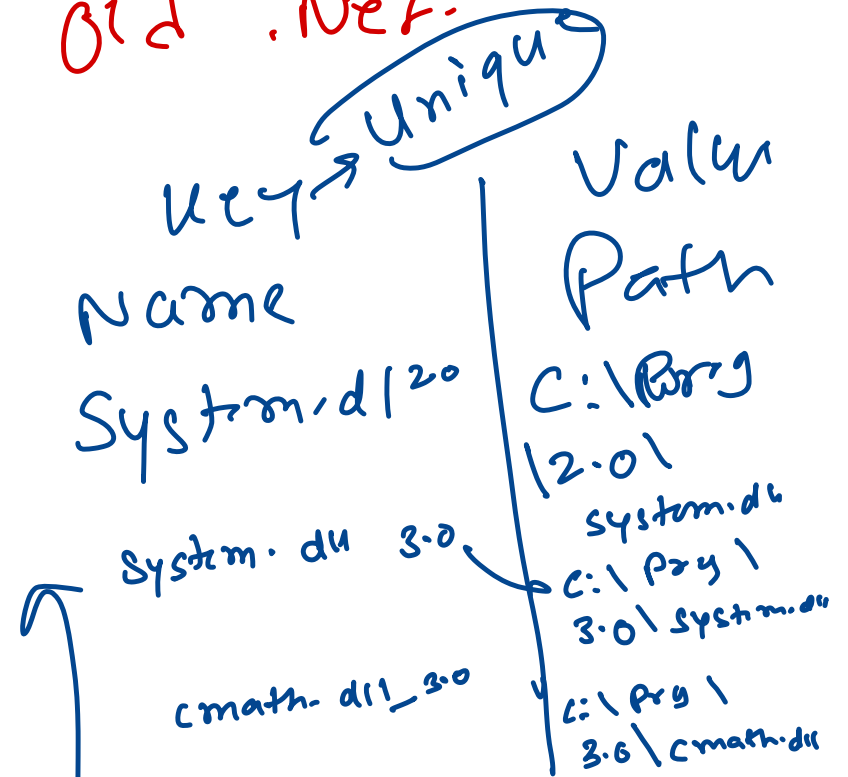
Private Assembly

↓
Convert

↓
Shared Assembly. ?

OS:- Windows

Old .NET.



Windows Registry / .NET Registry

GAC :- Global Assembly Cache.

All FCL by default gets registered after download / Install

Cmath $\Rightarrow$ CAC $\rightarrow$ register

$\downarrow$ assigning unique key

cmd: $\rightarrow$ gacuti Cmath.d1

$\downarrow$

cmd: SingIn

$\rightarrow$ unique key generat

$\rightarrow$ save $\rightarrow$ assembly.

calculator.exe → D:\MyApp\calculator.exe
→ D:\MyApp\cmath.dll X

Private Assembly

cmath.dll

Advance Calc.exe → D:\ABC\AdvCalc.exe
D:\ABC\cmath.dll X

Shared Assembly
c:\Prog\3.0\cmath.dll
c:\Prog\3.0\System.dll
GAC registered path.

Structure of .NET Assembly.

PE Header. call OS • call CLR (mscorlib.dll)
Assembly metadata
Resource metadata
Type metadata
MSIL

→ version 2.0, © BRS, security,
private key, -

→ audio, video, pdf, images.

→ class, interface, static, sealed,
abstract, delegate, ctor, data-types
namespace, properties, methods
virtual, override

Calculator.exe ~~↗~~ ← Portable Executable
main() → method ~~↗~~

OS : windows

Visual Studio 2022

~~★~~ community edition

~~★~~ .NET F/W

~~★~~ SQL Express : Free DB server

~~★~~ IIS Express : Web server

Netel Server : Web server